PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Clinical efficacy of adjuvant treatments for patients with resected biliary tract cancer: a systematic review and network meta-analysis
AUTHORS	Chen, Ye; Zhang, Baoxia; Liu, Chang; Cao, Ye; Lyu, Cheng; Qiu, Meng

VERSION 1 – REVIEW

REVIEWER	Alessandro Rizzo
REVIEW RETURNED	18-Sep-2021

GENERAL COMMENTS	Dear Editor, thank you so much for inviting me to revise this manuscript about adjuvant therapy in biliary tract cancer. This study addresses a current topic. The manuscript is quite well written and organized. English could be improved. Figures and tables are comprehensive and clear. The introduction explains in a clear and coherent manner the background of this study. We suggest the following modifications: • Introduction section: although the authors correctly included important papers in this setting, we believe a couple of studies should be cited within the introduction (PMID: 33307876; PMID: 31986437; PMID: 33571059), only for a matter of consistency. We think it might be useful to introduce the topic of this interesting study. • Methods and Statistical Analysis: nothing to add. • Discussion section: Very interesting and timely discussion. Of note, the authors should expand the Discussion section, including a more personal perspective to reflect on. For example, they could answer the following questions – in order to facilitate the understanding of this complex topic to readers: what potential does this study hold? What are the knowledge gaps and how do researchers tackle them? How do you see this area unfolding in the next 5 years? We think it would be extremely interesting for the readers. However, we think the authors should be acknowledged for their work. In fact, they correctly addressed an important topic, the methods sound good and their discussion is well balanced. One additional little flaw: the authors could better explain the limitations of their work, in the last part of the Discussion. We believe this article is suitable for publication in the journal although major revisions are needed. The main strengths of this paper are that it addresses an interesting and very timely question

	and provides a clear answer, with some limitations. We suggest a linguistic revision and the addition of some references for a matter of consistency. Moreover, the authors should better clarify some points.

REVIEWER	Héloïse Bourien Centre Eugene Marquis
REVIEW RETURNED	23-Sep-2021

GENERAL COMMENTS	This meta-analysis permits an update for adjuvant therapy in BTC. Methods: Some relevant studies were not in the analysis : Gold and al (2) … The meta-analysis seems not to be exhaustive. Results: - Supplementary Figure 1: is a little bit confused and is not well explained, remove it if this Figure is not necessary. - Figure 1: Could the authors explain why do they have excluded 15 studies with mixed adjuvant therapies? - Supplementary Figure 6: for D: I suggest to separate OS and DFS in two separate forest plots. Could you explain “non-distal metastasis”, is patient without metastasis? These data are important must be in the manuscript not in supplementary data. Discussion: Line 33: It is important to add “after BTC patients”, in randomized studies. In the discussion, the authors did not point that the unique positive randomized trial of adjuvant treatment in BTC is the BILCAP study. Whereas there is not randomized trial for chemo-radiation regimen or radiotherapy alone. The authors had to detail the heterogeneity of the population compared. The authors had to moderate their results, because of no direct comparison between studies.

REVIEWER	Hozaifa Khalil Elsawah Alexandria University High Institute of Public Health, Biostatistics
REVIEW RETURNED	06-Dec-2021

GENERAL COMMENTS	The network meta-analysis work is well organized, clearly presented, and updated. I recommend minor changes for better quality as the following: Avoid abbreviations in the abstract, and if necessary, it should be stated for the first mention time. In the PRISMA flow diagram, replace "the number of duplicates excluded" with "the number after duplicates excluded" to keep consistency of the flow chart. Check for errors in reporting numbers and groups in result section, because I found some such as in line 250; you stated "ACT had HR

	=0.74" but in figure 3B, it was ACRT. Another one is the confidence interval in line 252 (0.68 - 1.06) which differs from that in figure 3B (0.68 - 1.1) Check through out the results

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer 1:

Reviewer: 1

(1) Introduction section: although the authors correctly included important papers in this setting, we believe a couple of studies should be cited within the introduction (PMID: 33307876; PMID: 31986437; PMID: 33571059), only for a matter of consistency. We think it might be useful to introduce the topic of this interesting study.

Response:

Thanks a lot for the kind suggestions. We have added these papers in the Introduction at lines 82-92.

(2) Methods and Statistical Analysis: nothing to add.

Response:

Thanks for your support.

(3) Discussion section: Very interesting and timely discussion. Of note, the authors should expand the Discussion section, including a more personal perspective to reflect on. For example, they could answer the following questions – in order to facilitate the understanding of this complex topic to readers: what potential does this study hold? What are the knowledge gaps and how do researchers tackle them? How do you see this area unfolding in the next 5 years? We think it would be extremely interesting for the readers. However, we think the authors should be acknowledged for their work. In fact, they correctly addressed an important topic, the methods sound good and their discussion is well balanced.

Response:

Thanks a lot for your suggestion. We have largely rewritten the discussion by incorporating more personal perspectives and future perspectives. Please refer to the changes at lines 370-411/ Discussion.

(4) One additional little flaw: the authors could better explain the limitations of their work, in the last part of the Discussion. We believe this article is suitable for publication in the journal although major revisions are needed. The main strengths of this paper are that it addresses an interesting and very timely question and provides a clear answer, with some limitations.

Response:

The entire paragraph of limitation was rewritten in this manuscript. We hope our modifications make us explain the limitation of our work better. Please refer to the changes at lines 412-429/ Discussion.

(5) We suggest a linguistic revision and the addition of some references for a matter of consistency. Moreover, the authors should better clarify some points.

Response:

Thanks a lot for your careful review. We got help from a professional editing service company to do the proofreading for this revised manuscript. Changes in the main text are highlighted in yellow.

Response to Reviewer 2:

Reviewer: 2

(1) Some relevant studies were not in the analysis : Gold and al (2) …

The meta-analysis seems not to be exhaustive.

Response:

The paper published by Douglas Gold and colleagues, Adjuvant therapy for gallbladder carcinoma: the Mayo Clinic Experience (PMID: 19297105), was entered the second round of literature screening. The full article was carefully read and assessed. As stated in the Method, we have pre-specified eligibility criteria on the baseline characteristics; age, sex, disease severity need to be balanced between groups. In Gold's paper, the overall disease stage, N category, and T category were significantly different between the surgery-only group and the surgery and adjuvant treatment group. That is why this paper was excluded from our meta-analysis.

(2) Supplementary Figure 1: is a little bit confused and is not well explained, remove it if this Figure is not necessary.

Response:

Supplementary Figure 1 is removed.

(3) Figure 1: Could the authors explain why do they have excluded 15 studies with mixed adjuvant therapies?

Response:

The studies with mixed adjuvant therapies were excluded because the type of adjuvant therapy was not clearly documented. The network meta-analysis cannot be conducted properly by not knowing the data of patients with ACT, ART, or ACRT separately.

(4) Supplementary Figure 6: for D: I suggest to separate OS and DFS in two separate forest plots. Could you explain “non-distal metastasis”, is patient without metastasis?

These data are important must be in the manuscript not in supplementary data.

Response:

We have re-arranged the sequence of figures accordingly. Supplementary Figure 6 was moved to Figure 4. No distant metastasis means no spread from the original tumor to distant organs or distant lymph nodes has been found, which is equal to M0 in TNM staging.

Discussion:

(5) Line 33: It is important to add “after BTC patients”, in randomized studies.

Response:

Thanks for your careful review. We have rewritten the sentences in the discussion. The sentence you pointed out in line 33/ discussion had been deleted due to the updated results.

(6) In the discussion, the authors did not point that the unique positive randomized trial of adjuvant treatment in BTC is the BILCAP study. Whereas there is not randomized trial for chemo-radiation regimen or radiotherapy alone.

Response:

Thanks a lot for pointing out the uniqueness of the BILCAP study. We have added some discussion about the BILCAP study and RCTs in this field. Please refer to the changes at lines 385-392/ Discussion.

(7) The authors had to detail the heterogeneity of the population compared.

Response:

We have details of the population in Table 1. And based on the pre-specified eligibility criteria on the baseline characteristics, the major baseline data were balanced between the compared groups. Moreover, we have listed sources of heterogeneity for our meta-analysis in the limitations at lines 412-426/Discussion.

(8) The authors had to moderate their results, because of no direct comparison between studies.

Response:

Thanks a lot for your suggestion. We have updated our literature search, and the results of our meta-analysis are also updated accordingly. And we have interpreted our results more cautiously.

Response to Reviewer 3:

Reviewer: 3

(1) Avoid abbreviations in the abstract, and if necessary, it should be stated for the first mention time.

Response:

We have removed some abbreviations in the abstract. The abbreviation of hazard ratio and confidence interval seems necessary, and they are stated in their first appearance.

(2) In the PRISMA flow diagram, replace “the number of duplicates excluded” with “the number after duplicates excluded” to keep consistency of the flow chart.

Response:

We have modified the PRISMA flow diagram according to your advice.

(3) Check for errors in reporting numbers and groups in result section, because I found some such as in line 250; you stated “ACT had HR =0.74” but in figure 3B, it was ACRT.

Another one is the confidence interval in line 252 (0.68 - 1.06) which differs from that in figure 3B (0.68 - 1.1). Check through out the results.

Response:

We are sorry about this mistake. We have updated our literature search, and the results of our meta-analysis are also updated and checked accordingly. In addition, we also added some descriptions in figure legends to further clarify the data. Please refer to the changes at lines 745-747/ Figure legends.

VERSION 2 – REVIEW

REVIEWER	Alessandro Rizzo
REVIEW RETURNED	08-Feb-2022

GENERAL COMMENTS	The authors modified the manuscript according to our suggestions. We recommend Acceptance.

REVIEWER	Héloïse Bourien Centre Eugene Marquis
REVIEW RETURNED	23-Feb-2022

GENERAL COMMENTS	Abstract : The absence of prospective studies for ART and ACRT, had to be precised in the conclusion. Results of comparison between 3 regimens of adjuvant treatment through independent studies had to be described with more moderation. Authors had to discuss the heterogeneity of treatment modalities in each group of adjuvant treatment ; for example in ACT studies clinical outcomes were different in terms of chemotherapy regimens : fluoropyrimidines had shown efficacy (BILCAP) whereas gemcitabine not (Edeline, Eur J Cancer 2022).

VERSION 2 – AUTHOR RESPONSE

Response to Reviewer 1:

Reviewer: 1

Dr. Alessandro Rizzo

Comments to the Author:

(1) The authors modified the manuscript according to our suggestions.

We recommend Acceptance.

Response:

Many thanks for Dr. Alessandro Rizzo's appreciation of our work.

Response to Reviewer 2:

Reviewer: 2

Dr. Héloïse Bourien, Centre Eugene Marquis

Comments to the Author:

(1) Abstract :

The absence of prospective studies for ART and ACRT, had to be precised in the conclusion.

Response:

Thanks a lot for your suggestion. We have added following sentences in the Abstract "However, ART only showed improvement in OS. Due to the lack of head-to-head studies of ACT, ACRT and ART, the above results need to be further verified by prospective randomized controlled trials." Please refer to the changes at lines 42-45/Abstract.

(2) Results of comparison between 3 regimens of adjuvant treatment through independent studies had to be described with more moderation.

Response:

We have modified the descriptions in a modest way. Please refer to the changes in lines 37-39 & 42-45/Abstract, lines 290-292 and 323-324/Results, lines 384-385 and 401-403/Discussion, and 444-448/Conclusion.

(3) Authors had to discuss the heterogeneity of treatment modalities in each group of adjuvant treatment ; for example in ACT studies clinical outcomes were different in terms of chemotherapy regimens : fluoropyrimidines had shown efficacy (BILCAP) whereas gemcitabine not (Edeline, Eur J Cancer 2022).

Response:

Thanks for your careful review. We have added some discussion about the heterogeneity of treatment modalities as follows: "In addition, the comparisons between the different treatment modalities and treatment regimens may substantially contribute to heterogeneity among included studies. Patients in ACT, ART or ACRT groups were treated with different modalities. Even within the same treatment modality, different regimens offered various efficacy results. Such as, the adjuvant capecitabine monotherapy (BILCAP study) appeared more effective compared with observation, whereas adjuvant gemcitabine (BCAT study) or gemcitabine plus oxaliplatin (PRODIGE 12-ACCORD 18 study) did not.[98]" Please refer to the changes at lines 425-432/Discussion.